Name	Tony Su
Student ID	33809453
Studio Session	Studio 1

Part A: URL of the Visualisation

Walking the Uneven Map

https://tsuu0008.github.io/FIT3179_visualisation_2/

Part B: Domain Description

Describe your domain chosen for DV II

The visualisation explores walking opportunities in Victoria's official Recreation Tracks network. It helps readers understand where routes are located and how their distance, terrain, walking time and accessibility affect the choice of a walk.

Describe the Why and Who aspects of your DV II

Designed for everyday Victorian walkers, occasional explorers and families, the visualisation assumes no statistical or bushwalking expertise. It helps readers choose walks that suit their location, time, experience and companions' access needs.

Readers first choose an area, then compare the route's demands, time and access. Murrindindi has 32 recorded routes, but only seven record full or partial access; all seven are partial. Douglas Reserve Walk and Upper Yarra Walking Track illustrate why grade alone is insufficient: both are Grade 4, yet one is 400 metres long and the other is 95 km.

Describe the "What" aspect of your DV II

Victoria's Department of Energy, Environment and Climate Action publishes the Recreation Tracks dataset. The 23 October 2026 export contains 366 records; all 195 marked for walking are included, totalling 875.905 km. Attributes describe distance, grade, terrain, walking time, experience and accessibility. Twenty routes are marked FULL CLOSURE, so inclusion in the maps does not indicate current availability.

Population estimates at 30 June 2025 come from the Australian Bureau of Statistics Regional population 2024-25 dataset, Table 2, column D of the local government area workbook. Combining these with walking distance gives kilometres per 100,000 residents, allowing comparison across councils of different sizes. This measures recorded supply relative to population, rather than path use or crowding.

Official route coordinates and raw codes are preserved. C1-C5 are displayed as Grade 1-5, and numeric 3 as Grade 3. C6 is labelled Unclassified; missing grades are labelled Unknown. These labels have not been independently verified against public walking-track

grades. Walking times remain in published categories. Each route is counted once, in the council containing its length-weighted midpoint. Cross-boundary routes are not split. The year of the council boundaries remains unverified.

Route source: Victorian Government Recreation Tracks

Population source: ABS Regional population 2024-25

Grade-guide context: Parks Victoria

Describe the "How" aspect of your DV II

All 14 charts and maps use Vega or Vega-Lite. The overview supports pan, zoom and named-route selection to inspect actual paths; a regional map adds local detail. A choropleth compares total kilometres with population-adjusted supply. A proportional symbol map uses symbol area for distance and colour and shape for grade to support shortlisting. An equal-area grid map counts route midpoints in 25 km squares, revealing clusters that council totals can conceal.

A heatmap counts routes by grade and terrain to reveal common combinations. A stacked bar chart compares experience advice across terrain groups, helping readers consider preparation. A mosaic compares distance contributions rather than route counts: column width shows each terrain group's share of kilometres, while coloured block height shows its grade mix.

A lollipop chart shows the middle distance by grade; an individual-distance strip plot reveals variation and unusually long routes hidden by that summary. A parallel coordinates chart connects a selected route's grade, distance, duration and terrain for comparison. A bar chart compares route counts by walking time. A connected-dot chart is used for rank comparison, linking each council's population rank with its rank for walking kilometres per 100,000 residents.

A waffle chart shows how few rated routes record full or partial access: 15 of 180. The 8.3% share is rounded to eight coloured cells out of 100; 15 unrated routes are excluded. Partial access may cover only some sections. This gives readers a starting list for closer checking.

The single scrolling page uses consistent grade colours and reading guides for a general audience. Annotations explain findings and their significance; circled letters link callouts to the relevant marks. Filters update their own chart and annotations. The layout fits laptop screens and retains the original sketch's main sections, maps, grade guide and focus on short but demanding walks.

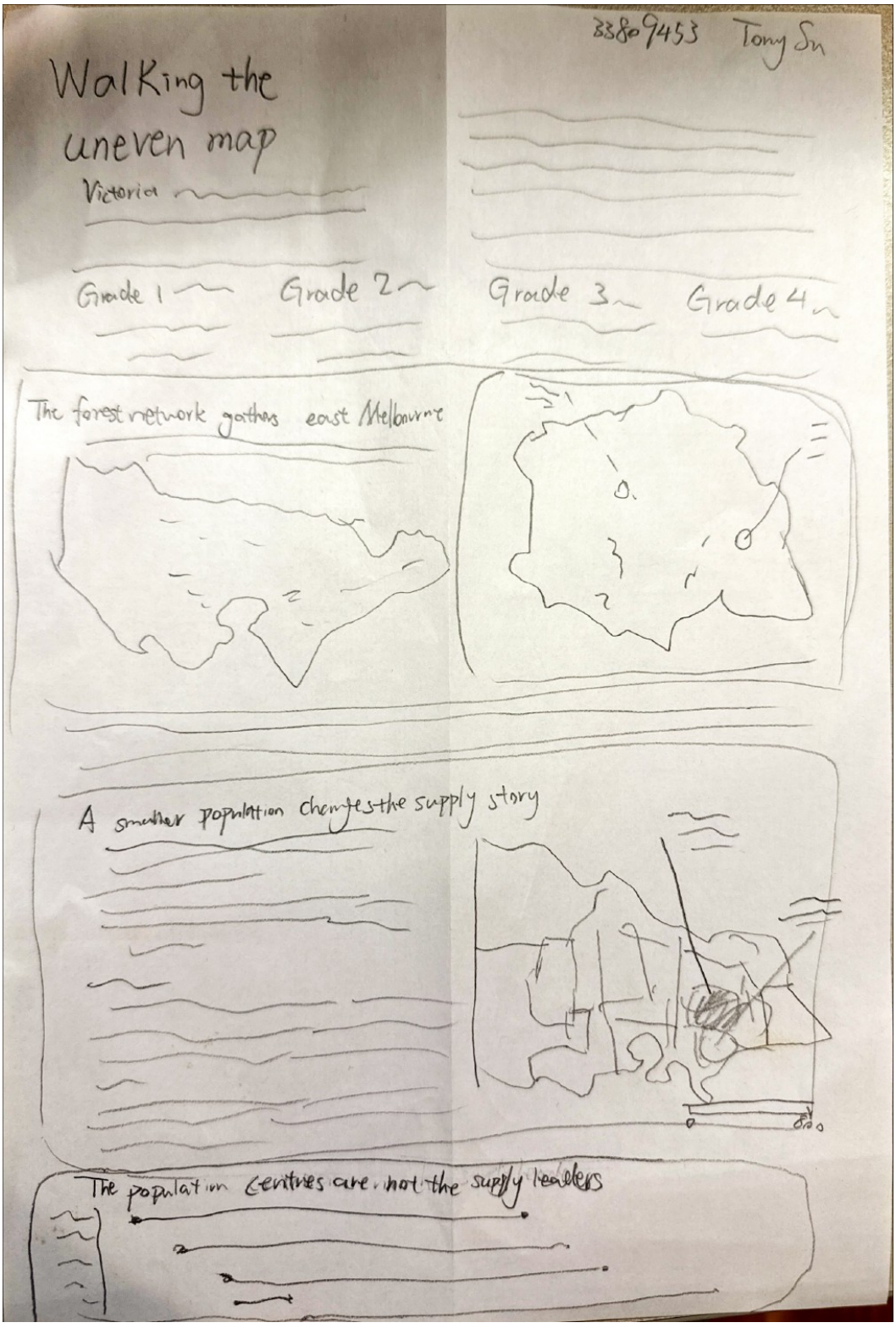
AI acknowledgement

I used OpenAI ChatGPT and Codex to assist with coding and to refine the wording of the website and this submission. I reviewed and revised the AI-assisted material and take responsibility for the final content, data interpretations and claims.

Part C: Sketch

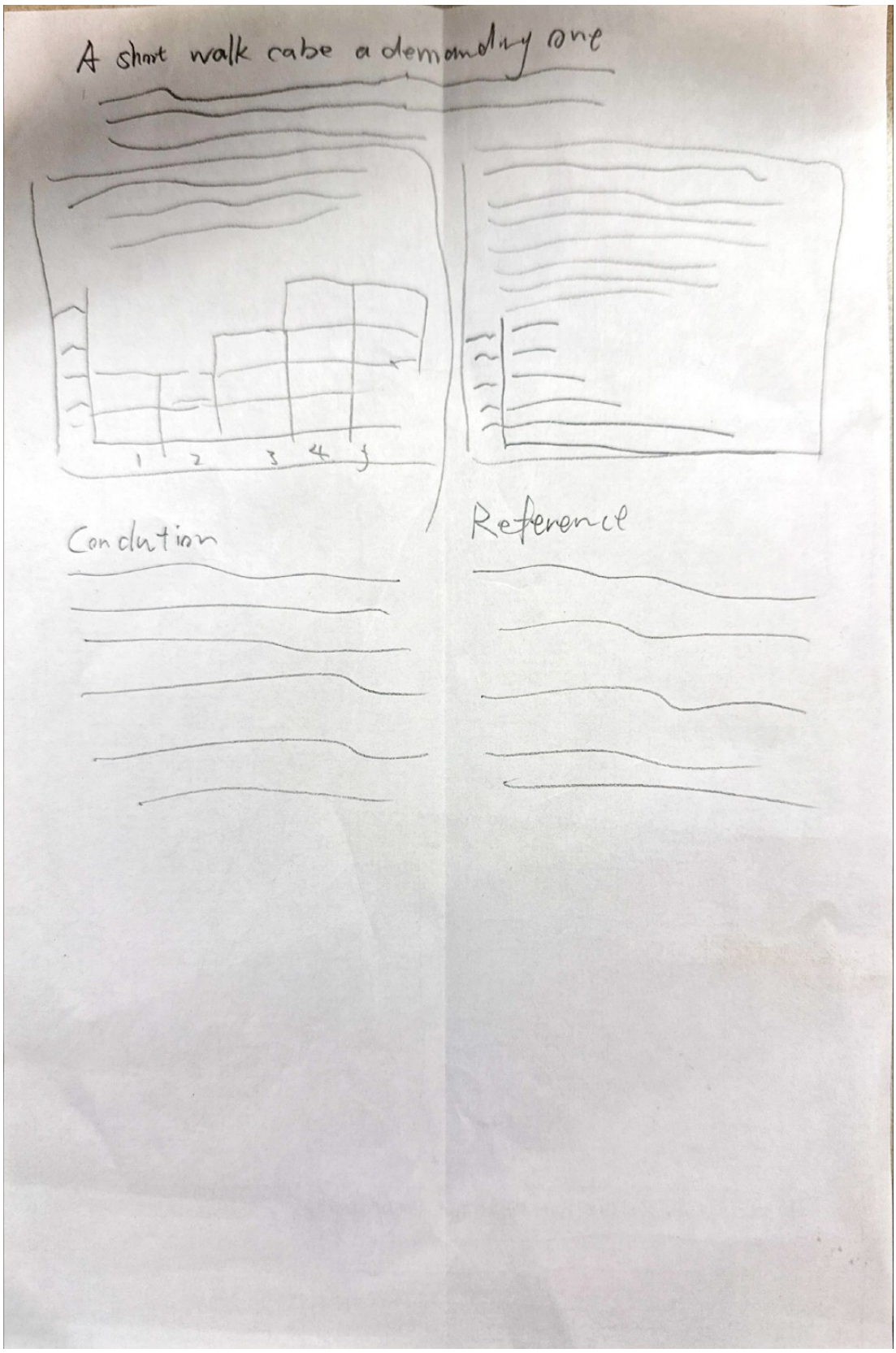
https://tsuu0008.github.io/FIT3179_visualisation_2/submission/DV2sketch.pdf

Original hand-drawn sketch, page 1 of 2. The original PDF is preserved unchanged.



Part C: Sketch continued

Original hand-drawn sketch, page 2 of 2.



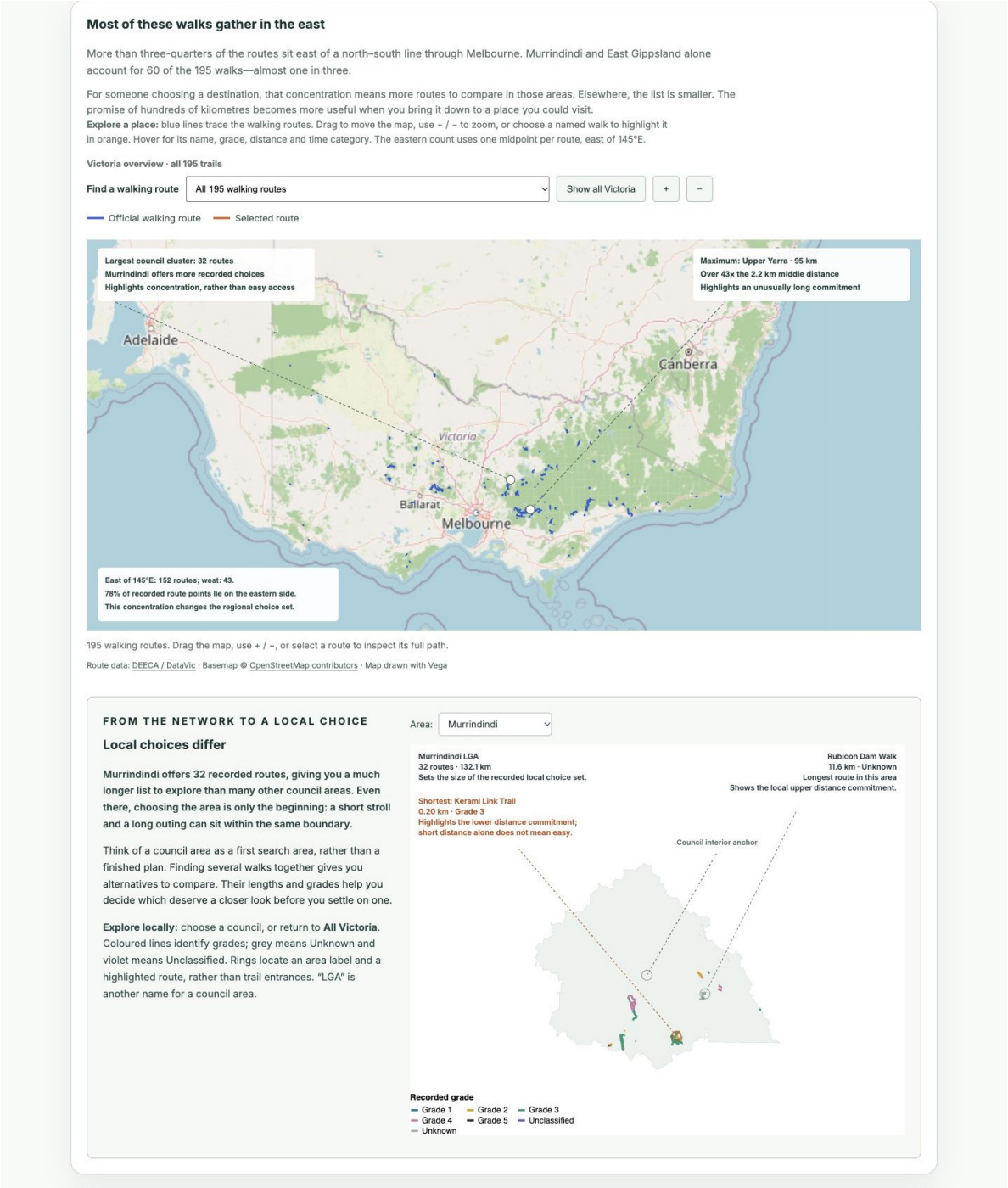
Part D: Full-size screenshot of your DV II

Walking the Uneven Map | 23 October 2026 | Segment 1 of 9



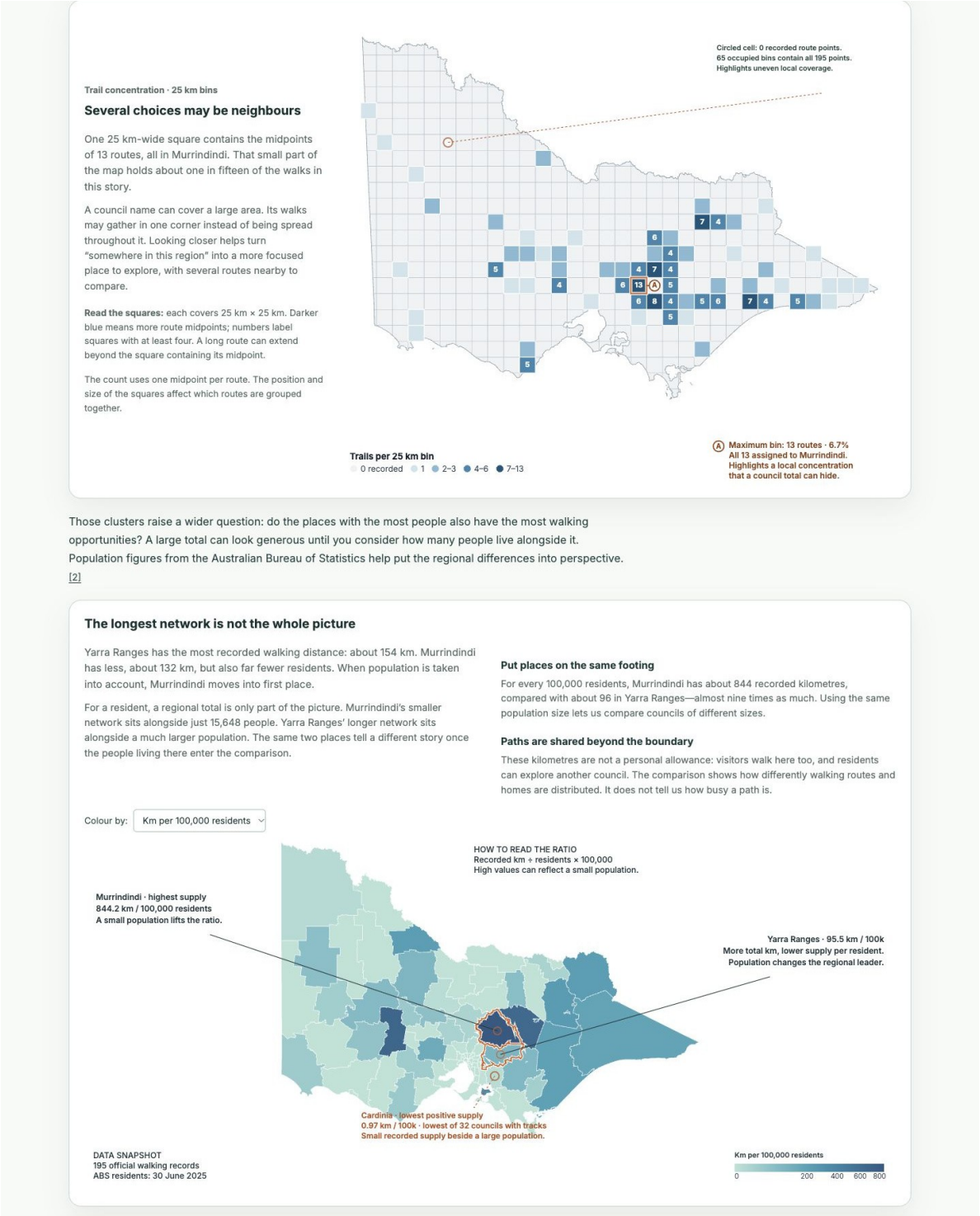
Part D: Website screenshot continued

Walking the Uneven Map | 23 October 2026 | Segment 2 of 9



Part D: Website screenshot continued

Walking the Uneven Map | 23 October 2026 | Segment 3 of 9



Those clusters raise a wider question: do the places with the most people also have the most walking opportunities? A large total can look generous until you consider how many people live alongside it. Population figures from the Australian Bureau of Statistics help put the regional differences into perspective. [2]

Part D: Website screenshot continued

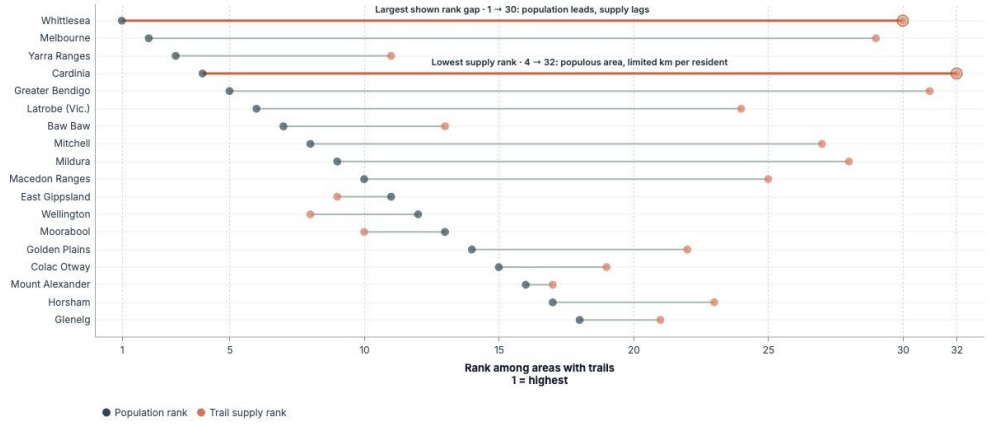
Walking the Uneven Map | 23 October 2026 | Segment 4 of 9

Where people live and paths gather do not always match

Whittlesea has the most residents among the 32 councils with these walking routes, yet comes 30th for walking kilometres relative to population. Cardinia is fourth for population and last for this measure.

A person living in a populous area cannot assume that this walking network offers a similarly large local choice. Murrindindi shows the reverse: a smaller population beside a relatively large network. Where people live and where these paths gather do not neatly match.

Read the positions: 1 means highest. Lines join each council's position for population and walking kilometres per 100,000 residents. The 18 most populous councils are shown; positions are calculated among all 32 with walking records. Line length shows a difference in position, not kilometres.



You now have a starting area, perhaps with several paths to explore. But a promising place is not yet a walk you would choose. The next decision happens along the path itself: will the ground, distance and experience needed suit you and your companions?

Next: being nearby does not make a walk suitable →

02 NEXT, LOOK BEYOND THE KILOMETRES

A short walk can still be a demanding one

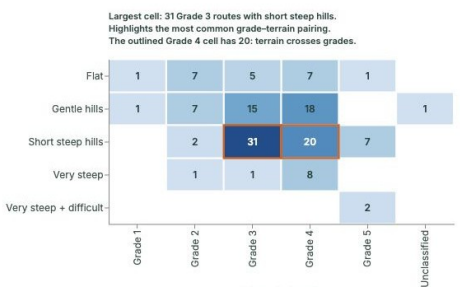
A short distance can be reassuring when you are planning an easy outing. Yet a few hundred metres of steep or rough ground can ask something very different of you from a level path. A walking grade is a useful first clue. It needs to be read alongside what the route says about its hills, surface and experience requirements.

One grade covers more than one kind of terrain

Short steep hills appear in both Grade 3 and Grade 4: 31 routes in one group and 20 in the other. A change in grade does not mean every feature of the ground changes with it.

If hills are the part of a walk that concerns you, the grade alone leaves a question unanswered. Two routes carrying the same grade can describe different ground; two carrying different grades can both include steep sections. Looking at the hills helps you choose around what matters to you.

Read the cells: rows describe the ground, columns show grades, and darker blue means more routes. There are 135 routes with both kinds of information. Missing grades are left out; Unclassified stays separate.

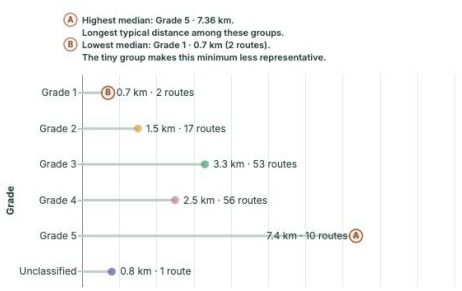


A higher grade is not simply a longer walk

The middle Grade 4 route is 2.5 km long, shorter than the middle Grade 3 route at 3.3 km. Moving up a grade does not simply add kilometres.

That matters when a shorter walk looks like the easier choice. Distance tells you how far you will go; the grade brings other demands into the decision. Reading both helps avoid mistaking a brief walk for a gentle one.

Read the points: the median is the middle distance when a group's routes are ordered from shortest to longest. Counts beside the points show how many routes are in each group. Grade 1 has only two routes and Unclassified only one, so their middle distances describe very small groups.



Part D: Website screenshot continued

Walking the Uneven Map | 23 October 2026 | Segment 5 of 9



Now bring that question down to an actual walk. Douglas Reserve Walk is just 400 metres, while Upper Yarra Walking Track stretches for 95 km. Both carry Grade 4 in these records. Sharing a grade clearly does not mean sharing the same kind of day out.

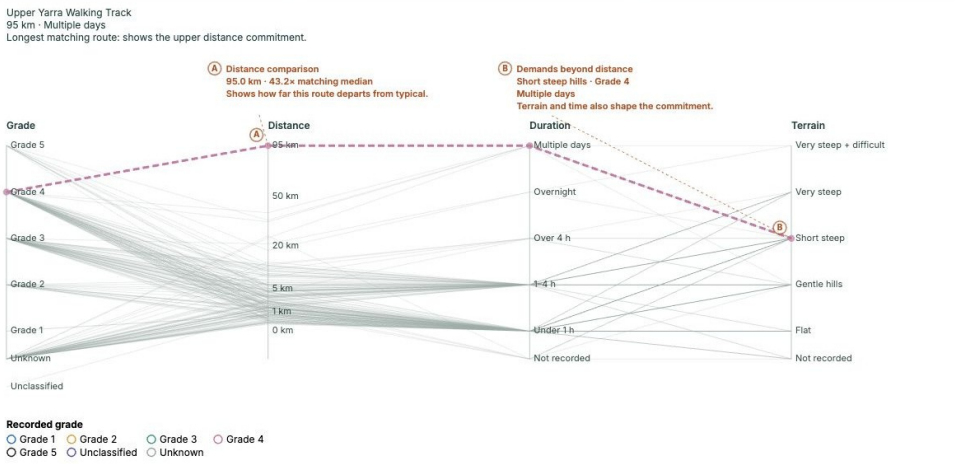
One grade, very different days out

Douglas Reserve Walk is listed as under an hour, with short steep hills and bushwalking experience recommended. Upper Yarra Walking Track is a multiple-day route for experienced bushwalkers. Both are Grade 4, but their lengths and time commitments are far apart.

The shorter option still asks you to consider hills and experience. The longer one asks for several days as well. This is where a route becomes more than a grade: you need to bring together how far it goes, how long it takes and what it asks of the people walking it.
Compare a walk: choose Douglas Reserve Walk or Upper Yarra Walking Track in the Trail menu. Follow the highlighted line across Grade, Distance, Duration and Terrain. The distance callout compares it with the median, or middle distance, of the matching walks. Long distances are compressed; use the kilometre labels. The other columns are categories, so their spacing does not measure effort.

One line, one walk: the thick dashed line is your selected route; faint lines show the other routes. Grade and Area menus narrow the choices before you select a trail.

Grade: All | Area: All | Trail: Longest matching



Unknown and Not recorded keep missing information visible. Unclassified is separate from Grade 1–5. The lines join features of a walk; they do not show a journey changing over time. Data and method

The short walk has not removed the need to think about experience. That is the next part of the choice: does a route welcome someone new to bushwalking, or does it ask for skills built on previous walks?

Experience becomes part of the choice

Of the 83 routes with short steep hills, 53 advise some bushwalking experience: 48 recommend it and another five call for experienced walkers.

For an occasional walker, that advice can change which route makes the shortlist. A distance you are comfortable with may still come with an experience requirement you did not expect. For a group, it is worth considering the person with the least experience as well as the keenest walker.
Read the bars: bar length counts routes, and coloured sections show recorded experience advice. Direct labels identify the larger segments. The comparison includes 188 routes with both terrain and experience recorded.

Hills are a substantial part of the choice

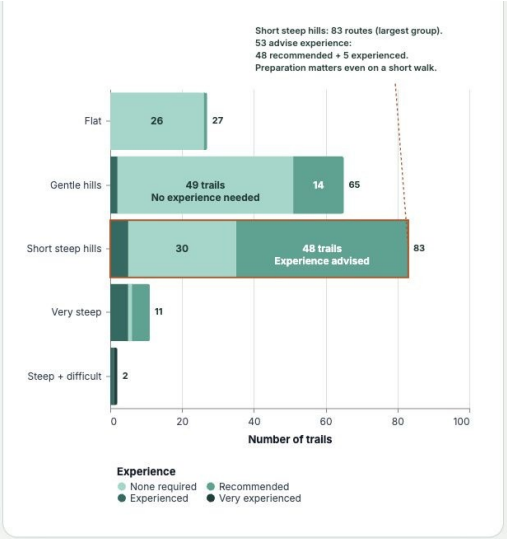
Routes described as having short steep hills contribute about 391 km—55% of the distance with both grade and terrain information. They are a substantial part of the walking on offer.

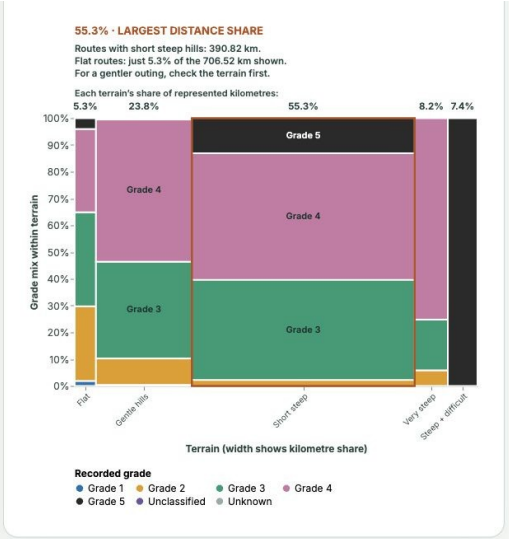
Someone looking for gentler ground is making a meaningful choice, rather than avoiding a rare exception. More than half the distance in this comparison belongs to routes described as having short steep hills. That describes the routes as a whole; it does not mean every metre of them is steep.
Read the rectangles: wider columns mean more walking kilometres in that terrain group. Colours show grades, and taller coloured blocks mean a larger share of that column's kilometres. Percentages above the columns show each terrain group's share of the distance.

This comparison covers 706.52 km with grade and terrain information. The full walking network totals about 876 km.

Part D: Website screenshot continued

Walking the Uneven Map | 23 October 2026 | Segment 6 of 9





You have moved from choosing a place to understanding the walk itself. A useful choice now has more than a name and a distance: its ground and experience advice make sense for you. The next step is to fit that walk into the time you have—and the needs of the people joining you.

Next: turn the network into a personal shortlist →

03 MAKE IT FIT YOUR DAY

Turn possibilities into a shortlist

Return to that free afternoon. You have somewhere in mind and a clearer idea of what you want underfoot. Now the choices have to work together: a manageable walk, enough time to enjoy it and the right conditions for the people coming along.

A region with many walks may offer you only a few

Return to Murrindindi, where the story began with 32 routes. If full or partial access is essential, that list becomes seven recorded possibilities. All seven are marked partial access.

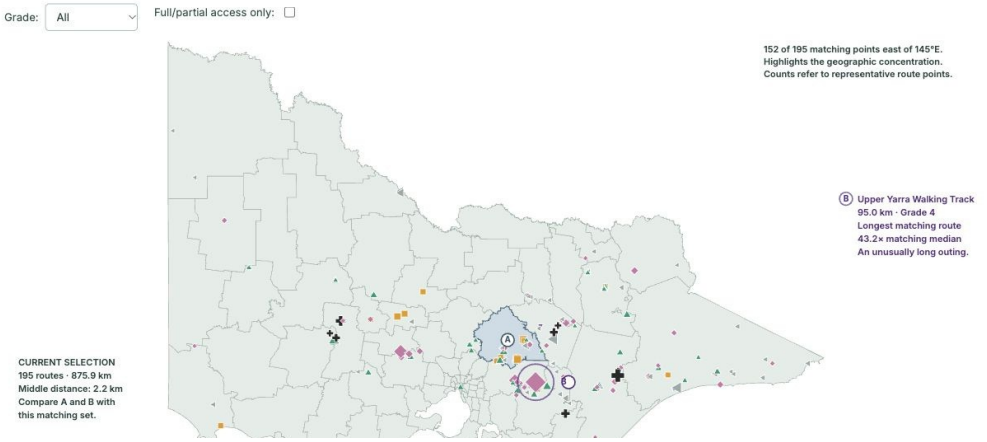
The region has not changed, but the choices relevant to your group have. A long list of walks is useful only when some fit your needs. Comparing grade, distance and access together brings the regional promise closer to a day you could actually plan. **Build a shortlist:** choose a grade and try the access filter. Larger symbols mean longer routes; colour and shape identify grade. Circled A marks the council with the most matches, and B the longest matching route. Symbols locate route midpoints, rather than entrances.

Think about everyone coming along

A companion who needs accessible facilities changes the starting question from "Which walk looks appealing?" to "Which walks describe the access we need?" Across Victoria, the filter keeps 15 routes marked full or partial access. The final chapter looks more closely at that small group.

A possibility still needs a closer look

Partial access may apply to only part of a route. Murrindindi's seven possibilities therefore need more detail before one becomes a plan. The access description matters alongside the distance and grade, rather than being a final yes-or-no answer.



Part D: Website screenshot continued

Walking the Uneven Map | 23 October 2026 | Segment 7 of 9

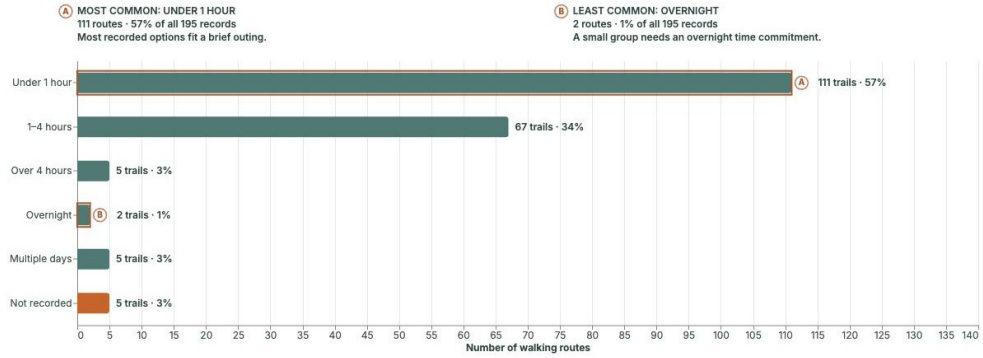


A brief outing has plenty of possibilities

More than half the walks—111 of 195—are listed as taking under an hour. Another 67 fall between one and four hours. Most belong to time categories that leave room for other plans in the day.

That is encouraging if you have an afternoon rather than a weekend. Yet a few routes ask for much more: five take over four hours, two are overnight and five span several days. Choosing a time category helps keep an appealing name on the map from becoming a bigger outing than you intended.

Read the bars: longer bars mean more walks in that time category. Labels show counts and percentages of all 195 routes. Five routes have no time recorded, shown in orange. Categories describe the published walking time, not an exact finish time or travel to the trail.

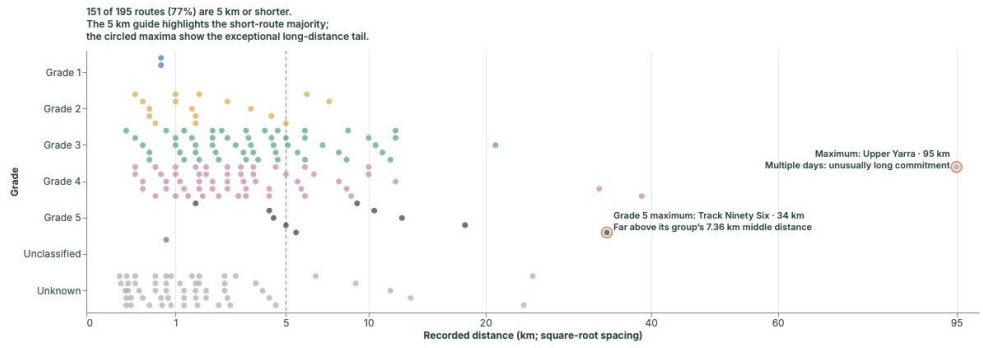


Most walks are short; a few change the scale

About three in four routes—151 of 195—are no longer than 5 km. The middle distance is just 2.2 km. The 95 km Upper Yarra Walking Track sits far beyond what most walks here ask in distance.

There is plenty to explore without choosing one of the longest routes. But remember the 400-metre Douglas Reserve Walk: short can still mean hills and experience advice. A distance that fits your afternoon becomes a useful choice only when the rest of the walk fits too.

Read the dots: each is one route. Long distances are compressed to give short walks more space; read the kilometre labels rather than judging by equal gaps. The dashed guide marks 5 km. Dots are nudged vertically to keep overlapping routes visible.



The afternoon walk is starting to take shape: somewhere you could visit, a distance and time you can allow, and demands that suit your group. One question still runs through the whole choice. Can everyone coming along use the path?

Next: a suitable distance still does not guarantee accessibility →

Part D: Website screenshot continued

Walking the Uneven Map | 23 October 2026 | Segment 8 of 9

04 MAKE ROOM FOR EVERYONE COMING ALONG

The choice has to work for everyone

An inviting walk can still leave someone out. For a person who needs accessible facilities, the important starting point is whether the route describes those facilities at all. Across the whole network, only 15 routes record full or partial access. The generous total of nearly 876 km now looks very different through that person's needs.

8.3%

15 of 180 rated routes record full or partial access.

The other 15 routes have no rating and are excluded from this percentage.

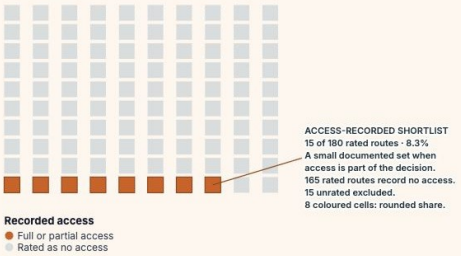
The ratings describe published access information. Partial access may cover only some sections or facilities.

A small starting list when access matters

Of the 180 routes with an access rating, 15 record full or partial access—about eight in every hundred. Another 15 routes have no rating and are left out of that percentage.

For someone who needs these features, access belongs near the start of choosing a walk. Finding a beautiful destination and a manageable distance is not enough if a companion cannot use the relevant section. The 15 routes offer a starting list for closer checking, rather than a promise that every one will suit every person.

Read the squares: imagine 100 routes with access ratings. About eight would record full or partial access, shown in orange. The actual count is 15 out of 180; the picture rounds 8.3% to eight squares.



For some walks, even the first questions remain open. Fourteen of the 15 routes without an access rating also lack a grade. Someone considering one of those routes has less help judging both its demands and its access. Clear information is part of making a choice possible.

WHAT THE STORY TELLS US

A usable walk has to fit a place, a day and a person.

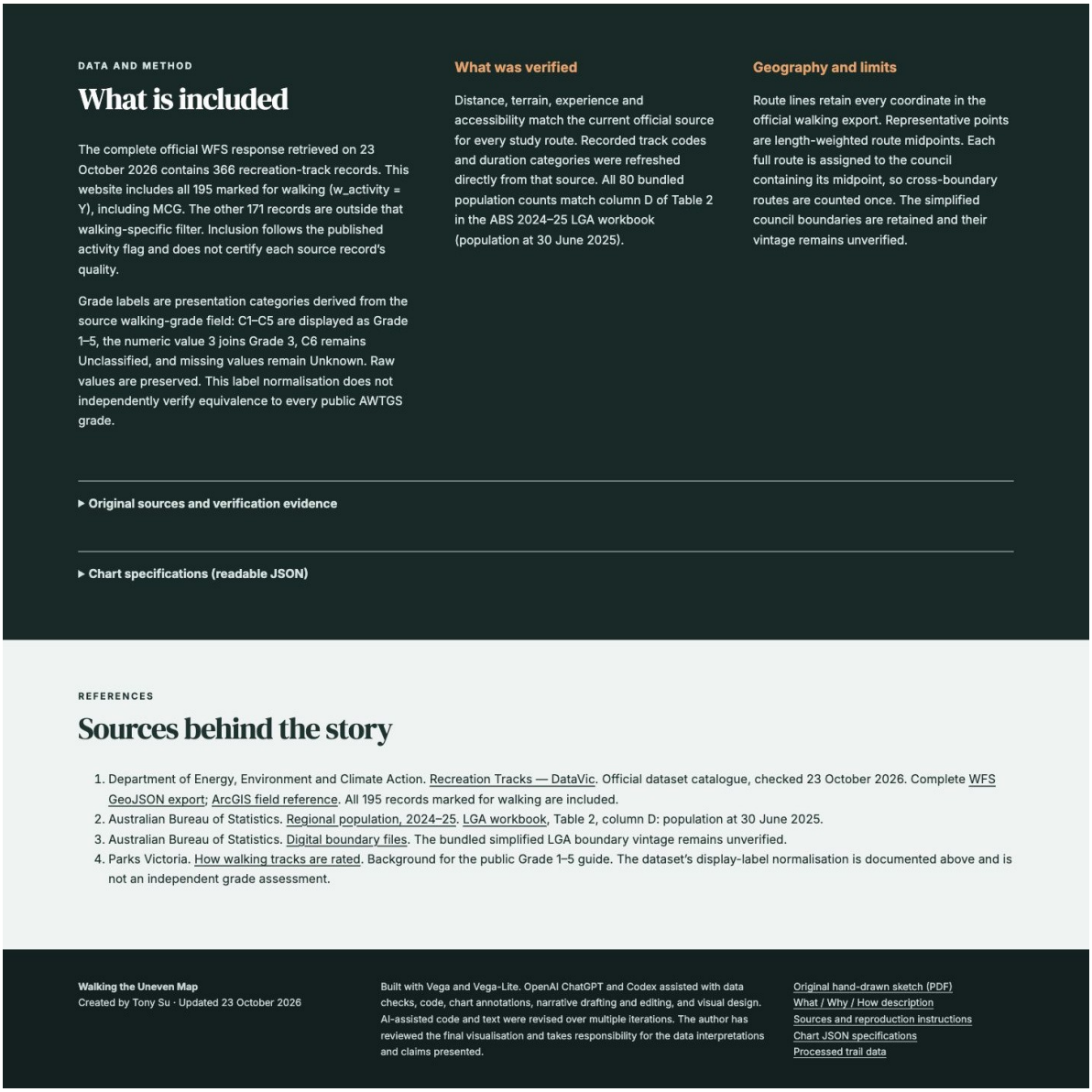
Return to the free afternoon that began this story. You started with 195 routes and nearly 876 kilometres. The useful choice emerged through smaller questions: where could you go, what would the ground ask of you, how much time could you give it, and would it work for everyone coming along?

Most walks here are short, so a modest outing has many possibilities. Those possibilities gather unevenly, and a short distance can still come with steep ground or a need for experience. Where accessible facilities matter, the documented list becomes much smaller again.

The best walk for your day is not necessarily the longest, the nearest or the one with the lowest grade. It is one whose place, demands, time and access fit together for your group. That is how a line on Victoria's uneven map becomes a walk you can begin to plan.

Part D: Website screenshot continued

Walking the Uneven Map | 23 October 2026 | Segment 9 of 9



DATA AND METHOD

What is included

The complete official WFS response retrieved on 23 October 2026 contains 366 recreation-track records. This website includes all 195 marked for walking (w_activity = Y), including MCG. The other 171 records are outside that walking-specific filter. Inclusion follows the published activity flag and does not certify each source record's quality.

Grade labels are presentation categories derived from the source walking-grade field: C1–C5 are displayed as Grade 1–5, the numeric value 3 joins Grade 3, C6 remains Unclassified, and missing values remain Unknown. Raw values are preserved. This label normalisation does not independently verify equivalence to every public AWTGS grade.

What was verified

Distance, terrain, experience and accessibility match the current official source for every study route. Recorded track codes and duration categories were refreshed directly from that source. All 80 bundled population counts match column D of Table 2 in the ABS 2024–25 LGA workbook (population at 30 June 2025).

Geography and limits

Route lines retain every coordinate in the official walking export. Representative points are length-weighted route midpoints. Each full route is assigned to the council containing its midpoint, so cross-boundary routes are counted once. The simplified council boundaries are retained and their vintage remains unverified.

▸ **Original sources and verification evidence**

▸ **Chart specifications (readable JSON)**

REFERENCES

Sources behind the story

1. Department of Energy, Environment and Climate Action. Recreation Tracks — DataVic. Official dataset catalogue, checked 23 October 2026. Complete WFS GeoJSON export; ArcGIS field reference. All 195 records marked for walking are included.
2. Australian Bureau of Statistics. Regional population, 2024–25. LGA workbook, Table 2, column D: population at 30 June 2025.
3. Australian Bureau of Statistics. Digital boundary files. The bundled simplified LGA boundary vintage remains unverified.
4. Parks Victoria. How walking tracks are rated. Background for the public Grade 1–5 guide. The dataset's display-label normalisation is documented above and is not an independent grade assessment.

Walking the Uneven Map
Created by Tony Su · Updated 23 October 2026

Built with Vega and Vega-Lite. OpenAI ChatGPT and Codex assisted with data checks, code, chart annotations, narrative drafting and editing, and visual design. AI-assisted code and text were revised over multiple iterations. The author has reviewed the final visualisation and takes responsibility for the data interpretations and claims presented.

Original hand-drawn sketch (PDF)
What / Why / How description
Sources and reproduction instructions
Chart JSON specifications
Processed trail data